

DIFFRACTION

27.1 DIFFRACTION

It is a matter of common experience that waves bend round obstacles placed in their path. For example water waves escaping through a small hole spread out in all directions as if they have originated at the hole. Similarly sound, form of wave motion, is found to pass round obstacles of moderate dimensions. The amount of bending, however, depends upon the size of the obstacle and the wavelength of wave. It was argued that light, which is a wave motion, should also bend round obstacles. A minute investigation reveals that light suffers some deviation from its straight path in passing close to the edges of opaque obstacles and narrow slits. The deviation is extremely small when the wavelength of light waves is small in comparison to the dimensions of the obstacle or aperture. But when the size of the aperture is comparable with the wavelength of light, this deviation becomes much pronounced. For example when light waves diverging from a narrow slit S (Fig. 27.1) which is illuminated by a mono-

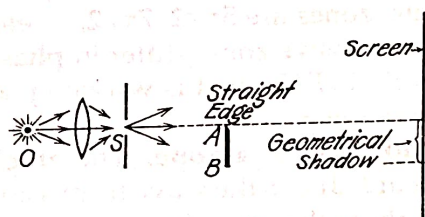


Fig. 27.1

chromatic source O , pass an obstacle AB with straight edge A parallel to the slit, the geometrical shadow on the screen is never sharp. A small portion of light bends around the edge into geometrical shadow. Outside the shadow parallel to its edge several bright and comparatively dark bands are observed.

Thus when light falls on obstacles or small apertures whose size is comparable with the wavelength of light, there is a departure from straight lines propagation, the light bends round the corners of the obstacles or apertures and enters in the geometrical shadow. This bending of light is called diffraction. It was found that diffraction produces bright and dark fringes known as diffraction bands or fringes.

The correct interpretation of diffraction phenomenon was provided by Fresnel. According to Fresnel, the diffraction phenomenon is due to mutual interference of secondary wavelets originating from various points of the wavefront which are not blocked off by the obstacle. Fresnel, thus, applied Huygens' principle of secondary wavelets in conjunction with the principle of interference and calculated the position of fringes. It was observed that the calculated results were in good agreement with the observed diffraction pattern. It should be remembered that the diffraction effects are observed only when a portion of the wavefront is cut off by some obstacle.

27.2 TWO KINDS OF DIFFRACTION

Diffraction phenomenon can be divided into following two general classes :

(1) **Fresnel's diffraction.** In this class of diffraction, source and screen are placed at finite distances from the aperture of obstacle having sharp edges. In this case no lenses are used for making the rays parallel or convergent. The incident wavefront are either spherical or cylindrical.

(2) **Fraunhofer's diffraction.** In this class of diffraction source and the screen or telescope (through which the image is viewed) are placed at infinity or effectively at infinity. In this case the wavefront which is incident on the aperture or obstacle is plane.

27.3 DIFFERENCE BETWEEN INTERFERENCE AND DIFFRACTION

Following are the differences between interference and diffraction phenomena :

1. In the phenomenon of interference, the interaction takes place between two separate wavefronts originating from the two coherent sources while in the phenomenon of diffraction the interaction takes place between the secondary wavelets originating from different points of the exposed parts of the same wavefront.

2. In the interference pattern the regions of minimum intensity are usually almost perfectly dark while it is not so in diffraction pattern.

3. The widths of the fringes in interference may or may not be equal or uniform while in diffraction pattern fringe width of various fringes are never equal.

4. In an interference pattern all the maxima are of same intensity but in a diffraction pattern they are of varying intensity.

RESOLVING POWER OF OPTICAL INSTRUMENTS

28.1 MEANING OF RESOLVING POWER

When the two objects are very near to each other or they are at very large distance from our eye, the eye may not be able to see them as separate. If we want to see them separate, optical instruments such as telescope, microscope etc. for close objects and prism and grating etc. for spectral lines are employed. Even if we assume that the instruments employed are completely free from all optical defects, the image of a point object or line is not simply a point or line but it is a diffraction pattern with a bright central maximum and other secondary maxima having minima in between of rapidly decreasing intensity. Thus an optical instrument is said to be able to resolve two point objects if the corresponding diffraction patterns are distinguishable from each other. The ability of the instrument to produce their separate patterns is known as resolving power.

tinct point of zero intensity in between the two. Hence the two wavelengths are *well resolved*.

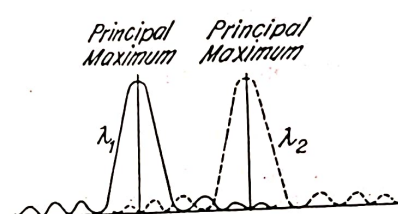


Fig. 28.1 (a)

Now consider the case when the difference in wavelengths is smaller and such that the central maximum of wavelengths coincides with the first minimum of the other as shown in Fig. 28.1 (b). The

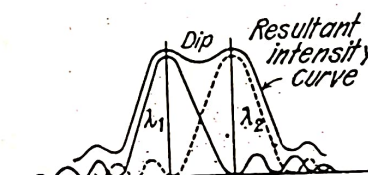


Fig. 28.1 (b)

resultant intensity curve is shown by thick curve. The curve shows a distinct dip in the middle of two central maxima, i.e., there is a noticeable decrease in intensity between the two central maxima indicating the presence of two different wavelengths. Thus the two wavelengths can be distinguished from one another and according to Rayleigh they are said to *just resolved*.

28.2 RAYLEIGH'S CRITERION OF RESOLUTION

According to Rayleigh criterion, two point sources are *resolvable* by an optical instrument when the central maximum in the diffraction pattern of one falls over the first minimum in the diffraction pattern of the other and *vice versa*. Similarly in case of spectral lines of two different wavelengths, the lines will be resolved when the central maximum due to one wavelength falls over the first minimum due to other and *vice versa*.

In order to illustrate the criterion let us consider the resolution of two wavelengths λ_1 and λ_2 by a grating. Fig. 28.1 (a) shows the intensity curves of the diffraction patterns of two wavelengths. The difference in wavelengths is such that their principal maxima are separately visible. There is a dis-

Again consider the case when the difference in wavelengths is so small that the central maxima corresponding to two wavelengths come still closer as shown in Fig. 28.1 (c). The resultant intensity curve in this case is quite smooth without any dip thus giving the impression as if there is only one wavelength source although somewhat bigger

28.2

and stronger. Hence the two wavelengths are *not resolved*.

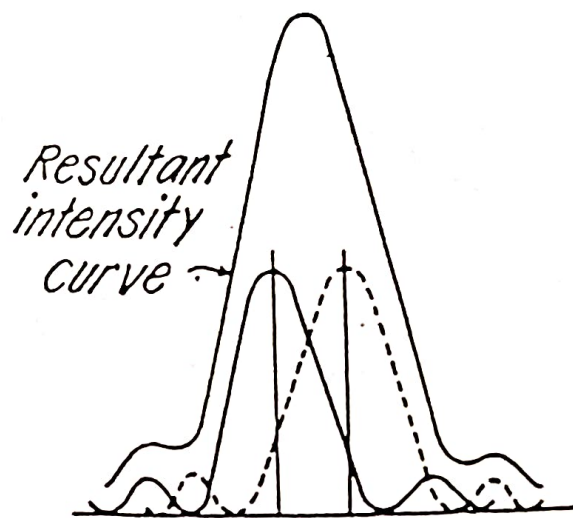


Fig. 28.1 (c)

Thus the two spectral lines can be resolved only upto a certain limit expressed by Rayleigh criterion.